OBSTACLE TRAINING

4

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OBSTACLE TRAINING (OT)

CHAPTER OT I : STANDARD OBSTACLE TRAINING

" Don't let the obstacles in your path be the reason you give up. Let them be the reason you push harder." - Unknown



TEACHING INSTRUCTIONS

Type	:	Lecture and Practice
Year	:	1st, 2nd & 3rd Year SD/SW
Conducting Officer	:	Permanent Instructor and ANO
Training Aids	:	Class room, OHP, Board, Chalk/Markers & OT Area
		SD/SW
Distribution of Periods	:	1st year - 3 Periods (1period theory and familiarisation 2 periods Practical on ground). 2nd year - 2 periods (Familiarisation and OT practice on ground) 3rd year - 2 periods (Familiarisation and OT practice on ground)

INTRODUCTION

1. The National Cadet Corps (NCC) stands as one of India's foremost youth organizations, dedicated to instilling character, leadership, and discipline in its cadets. Central to its comprehensive training regimen is the focus on both physical fitness and mental resilience, essential qualities for shaping the leaders of tomorrow. Among its diverse training modules, obstacle course training holds particular significance, as it sharpens not only the cadets' physical strength but also their problem-solving abilities, teamwork, and leadership skills. This training builds confidence, cultivates courage, and strengthens willpower, empowering cadets to overcome any challenge or barrier in life. It equips them to tackle both physical and mental obstacles, fostering resilience that will benefit them in all facets of their personal and professional journeys.

PART I : OBSTACLE COURSE AND METHODS OF NEGOTIATION

2. Obstacle training in the NCC is a rigorous and dynamic activity that blends physical exertion with mental resilience. Cadets must navigate a series of ten obstacles, each designed to teach proper techniques in jumping, balancing, scaling, and vaulting. This training plays a crucial role in fostering early fitness development among cadets. Every obstacle targets specific attributes such as strength, endurance, agility, coordination, and balance. The sequence of obstacles challenges cadets to think quickly, manage their energy efficiently, and maintain focus under pressure. The demanding nature of the course pushes cadets beyond their comfort zones, helping them uncover their true potential. The Standard Obstacle Course that NCC cadets must navigate comprises ten distinct obstacles, each spaced about 30 feet (10 yards) apart. Constructed from materials such as wood, bricks, concrete, and mud, these obstacles challenge cadets both physically and mentally. The course begins at the start line, and the cadets proceed to tackle obstacles in the following order: straight balance, clear jump, gate vault, zig-zag balance, high wall, double ditch, right vault, left vault, ramp, another straight balance, before finally crossing the finish line.

3. **Straight Balance**

(a) **Description**. The obstacle consists of a wooden slab measuring 3 inches thick, 4 inches wide, and 12 feet long, positioned 1.5 feet above ground level.

(b) **Method**. The cadet crosses the slab by running with arms outstretched to maintain balance. This obstacle emphasizes body balance and coordination. Relaxation is key to successfully navigating obstacles where balance is crucial. A relaxed approach builds confidence and enables faster crossings, while tension often results in loss of balance or, at best, slower progress. The position of the head plays a vital role in maintaining balance and should be kept as upright as possible, especially when moving on narrow surfaces or at heights. The eyes should neither point directly downward which may cause instability and impose more caution, nor look straight ahead which can lead to missteps or chances of fall. Ideally, the focus should be about 3 meters ahead. Since rifles are carried during obstacle courses, it is important to learn crossing the obstacle without raising of arms sideways for balance.





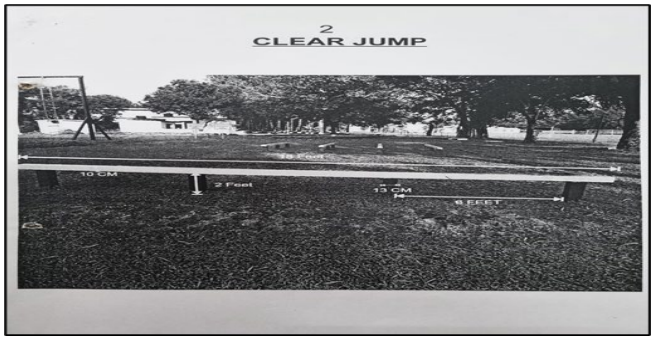




4. **Clear Jump**

(a) **Description**. Its structure resembles a straight wooden bar, 18 feet long and positioned 2 feet above the ground. The cadet must jump over the bar without touching it or using any part of the body for support.

(b) **Method**. After a short run toward the obstacle at a right angle, the cadet takes off from one foot. Both knees are then quickly raised high in front of the chest to ensure the feet clear the bar. To aid in lifting the body, the arms, slightly bent, swing forward as the knees rise. Once the obstacle is cleared, the knees and body extend, and the arms lower. The landing is executed by stepping off with one foot while moving the other just past the edge of the bar, jumping downward and slightly forward. Controlling body balance during both the flight and landing is key. The landing should be with feet and knees together, pointed forward, and with enough knee bend to absorb the impact, preventing jarring to the body







5. **Gate Vault**.

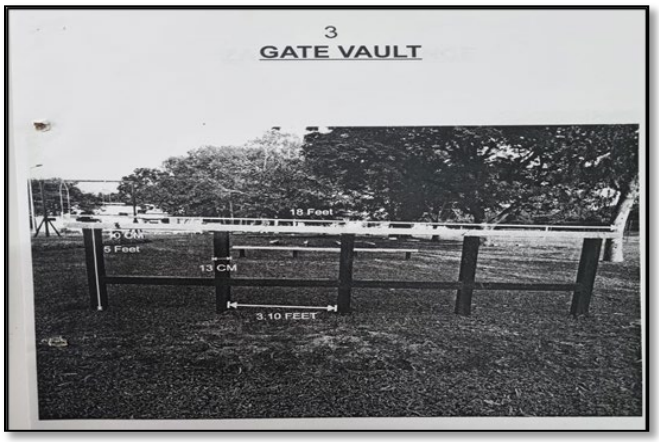
(a) **Description**. This obstacle consists of a wooden structure with two horizontal beams, positioned at heights of 3 feet and 5 feet, each 18 feet long.

(b) **Method**. To cross this gate, you must grip the upper beam with both hands, place your feet on the lower beam, and then jump across. This vaulting technique is particularly useful for overcoming gates. Begin with a short run-up, then take off from one leg, aiming to grip the top beam with both hands while jumping for the lower beam. Once you achieve balance, lean your torso forward and downward, moving one hand across to the opposite side of the structure until it aligns with the other hand. At this point, grasp the lower beam and roll your body over it. Common Faults in crossing gate vault is when cadet jumps instead of step up. Stepping up to the lower beam rather than jumping for it can result in a loss of speed and an uncontrolled landing.

(c) **Common Faults.**

(i) Stepping up instead of jumping up to the lower beam, which can result in loss of speed and lead to uncontrolled landing.

(ii) Keeping upper body balance back, which can lead to backward fall







6. **Zig-Zag Balance**.

(a) **Description**. An 18-foot-long wooden beam, constructed in a zig-zag manner, 3 inches wide and 1½ feet above the ground.

(b) **Method**. The cadet crosses the beam while balancing the body in a zig-zag motion, with arms open for support.

(i) **Approach the Obstacle**. Begin with a steady, controlled pace as you approach the zig-zag balance beam. Maintain a relaxed posture and focus on balance.

(ii) **Initial Step**. Step onto the beam confidently with your dominant foot. Keep your arms outstretched to the sides for balance.

(iii) **Foot Placement**. Place your feet carefully, one in front of the other, ensuring each step is centered on the beam. Avoid long strides, as shorter, more controlled steps offer better balance.

(iv) **Zig-Zag Transition**. As you reach each turn in the beam, pivot smoothly on the heels of your feet. Maintain control and avoid rushing through the turns, as this is where balance is most easily lost.

(v) **Maintain Focus**. Keep your head upright and focus your eyes a few feet ahead of each step, rather than looking directly down at the beam. This helps with overall balance and coordination.

(vi) **Arm Control**. Keep your arms slightly raised and extended for stability but avoid exaggerated movements. Subtle adjustments help maintain balance throughout the course.

(vii) **Finish Strong**. As you approach the end of the zig-zag beam, maintain your focus and continue with the same controlled steps. Step off the beam confidently once you reach the finish.









7. **High Wall**

(a) **Description**. The obstacle consists of a 6-foot-high, 12-foot-long bricked wall, plastered on both sides.

(b) **Method**. All cadets should try and cross this obstacle individually. For team exercises and for cadets who aren't able to cross it on its own, the "stirrup lift" team approach may be used.

In the **individual method**, the cadet runs towards the wall and leaps, kicking off the wall with one leg. Using both hands to grip the top, the cadet pushes upward, pulling their body onto the wall. Once gripping the top with both hands, the cadet bends their arms, first placing one forearm and then the opposite palm on the wall to gain leverage. This allows the cadet to roll over the top safely, rather than standing or sitting on it, minimizing exposure as also wasting time. In the **stirrup lift method**, used when teamwork is involved, two cadets stand against the wall facing each other. They form a "stirrup" by placing their hands on one cadet's bent leg, creating a platform for the climber. The climber steps into the stirrup and reaches for the top of the wall. As the lifters push up under his heels, the climber is propelled high enough to grasp the top and roll over. Like in the individual method, cadets should avoid sitting on the wall to prevent making themselves visible. Cadets are also trained in the correct descent method. After rolling over, they should hang down from the wall with fully extended arms, releasing one hand for balance. A strong push-off from the remaining hand should be made, turning their body so they land with their back to the wall, ensuring a safer landing

(c) **Common Faults**.

(i) Approaching the wall with speed and abruptly stopping owing to miscalculation resulting in injury.

(ii) Standing or taking a high position on the wall instead of rolling over







8. **Double Ditch**.

(a) **Description**. The obstacle consists of two ditches, each measuring 6-8 feet in length, 4-5 feet in width, and 3-4 feet in depth, with a small gap of 9-12 inches between them.

(b) **Method**. Begin with a preliminary run-up and take off from one foot. As you leap across the first ditch, place one foot on the small gap before jumping over the second ditch and landing across on the other foot. This maneuver requires precise agility and timing.

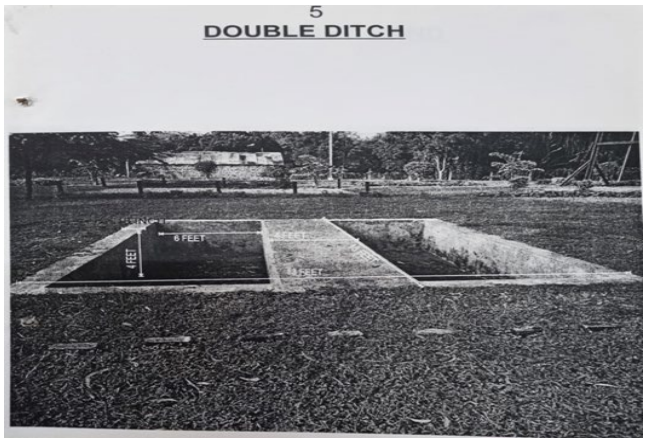
(c) **Common Faults are**.

(i) **Insufficient Speed**. A lack of speed during the run-up can hinder performance.

(ii) **Uncontrolled Flight**. Poor control during flight can affect your ability to clear the ditches.

(iii) **Lack of Height**. Inadequate height can lead to difficulties in clearing the obstacles.

(iv) **Uncontrolled Landing**. Landing on both feet instead of one, either in between the ditches or after the second one, will lead to body falling forward because of speed of motion. Ensure stable take off and landing on one foot









9. **Right Hand Vault.**

(a) **Description**. This wooden structure stands 3.5 feet above the ground and is 18 feet long. The cadet must jump over it by using the right hand for support while lifting both legs over the beam.

(b) **Method**. As far as possible, approach the obstacle straight, not from the side, and take off on one foot to cross the obstacle, while placing the right hand on top of the beam for support. The body's weight is shifted entirely onto the right arm as the cadet moves over the obstacle with the right side facing it. The landing be made on one foot to maintain momentum. Only right hand should touch the beam, violation of which incurs a penalty in competitions including TSC.







10. **Left Hand Vault**.

(a) **Description**. This wooden structure, similar to the Right Hand Vault, stands 3.5 feet above the ground and is 18.5 feet long.

(b) **Method**. As far as possible, approach the obstacle straight, not from the side, and take off on one foot to cross the obstacle, while placing the right hand on top of the beam for support. The body weight is shifted entirely onto the right arm as the cadet moves over the obstacle with the right side facing it. The landing be made on one foot to maintain momentum. Only left hand should touch the beam, violation of which incurs a penalty in competitions including TSC.

(c) **Common Faults.** Not maintaining enough momentum or stopping abruptly before crossing the obstacle, leading to injury.







HIGHER ORDER THINKING SKILLS (HOTS)

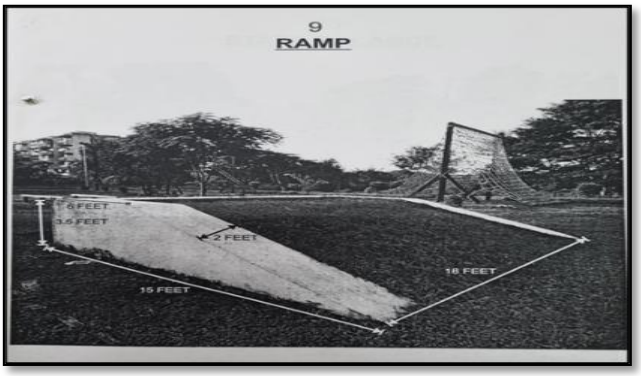
- What is the aim of obstacle course training and how many obstacles are included in the Standard Obstacle Training?
- How does the obstacle course training contribute to the development of NCC cadets?

11. **Ramp**.

(a) **Description**. The obstacle is a sloped hillock, measuring 15 feet in length, 18 feet in width, and 4.5 feet in height.

(b) **Method**. To cross it, the cadet runs up the slope to the top and takes a long jump, landing in the designated area, preferably on both feet. The knees have to be slightly bent to absorb the landing shock. The run-up should be steady and controlled, ensuring a smooth jump from the top and a balanced landing.

(c) **Common Faults.** Most injuries in this obstacle occur while landing. Major reason for the same is unsteady landing, especially on one foot, or landing without bending the knees even on both feet. Over weight participants are more at risk.



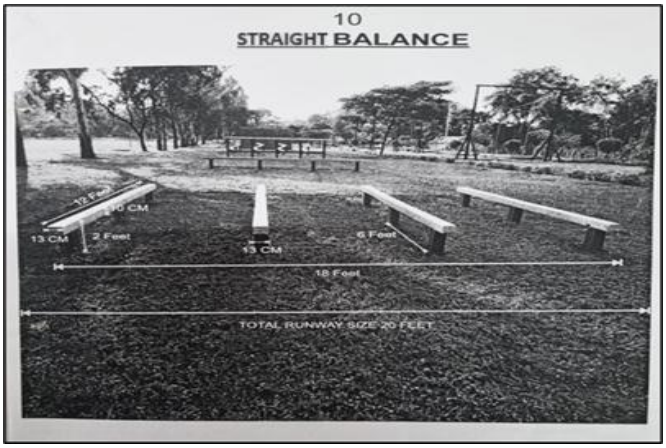




12. **Straight Balance**.

(a) **Description.** Similar to the first Straight Balance obstacle.

(b) **Method**: To approached as explained in first obstacle





PART II : DRESS & EQUIPMENT

13. **Dress**. The training attire should progress through different stages. Initially, obstacle-negotiating techniques should be taught in tracksuits or **Single belt** for the first two sessions. Belt is advised to be worn even over tracksuits to maintain taut posture. **Even when wearing tracksuits, only DMS boots will be worn. The obstacle course should**

not be done in PT shoes, sports shoes or Jungle boots, which may lead to severe injury. By the third session, cadets should wear disruptive shirts, trousers, and boots. In the final stage of instruction, typically during the Standard Obstacle Course, the attire will be Battle Order less rifle.

14. **Obstacles for Senior Wing (SW) Cadets**. Senior Wing girl cadets are required to navigate the following obstacles:-

(a) Straight Balance.

(b) Clear Jump.

(c) Gate Vault.

(d) Zig-Zag Balance.

(e) Ramp.

(f) Straight Balance (Repeated).

PART III : METHODOLOGY OF NEGOTIATING THE OBSTACLE COURSE

15. **Practice in Negotiating the Obstacle Course**. Regular practice is essential for mastering the skills needed to effectively navigate an obstacle course. Repeated exposure helps cadets build confidence, physical strength, agility, and mental resilience, all of which are key to overcoming various obstacles. During practice sessions, cadets should focus on improving their technique for each individual obstacle, such as jumps, vaults, balance beams, and climbing structures. Key aspects of successful obstacle course practice include:-

(a) **Progressive Training**. Start with basic techniques, gradually moving to more complex challenges as cadets build strength and skill.

(b) **Repetition for Muscle Memory**. Repeating each obstacle helps engrain proper technique, ensuring that movements become automatic and fluid during real-time challenges.

(c) **Attention to Form**. Emphasis should be placed on proper posture, balance, and footwork to prevent injury and maximize efficiency.

(d) **Developing Speed and Endurance**. As cadets gain proficiency, they should work on increasing their speed while maintaining control and precision. Endurance training is equally important for managing fatigue.

(e) **Mental Preparation**. Tackling obstacles often requires quick decision-making and adaptability, so practice sessions should also focus on building mental toughness and focus. With consistent practice, cadets improve both their physical capabilities and the confidence needed to negotiate the obstacle course effectively.

16. **Safety Measures for Successful and Safe Completion of the Obstacle Course**.

(a) **Participant Selection**. Only physically fit cadets should engage in obstacle course training.

(b) **Progressive Training**. Begin training in single belt, gradually progressing to more advanced stages with packs and weapons.

(c) **Warm up**. Proper warming up as per weather conditions is a must to practice safely and avoid injuries. Participants should only be allowed to enter the obstacle arena after having warmed up, which will include both running and stretching.

(d) **Avoid Hazardous Conditions**. Steer clear of wet or slippery surfaces to minimize the risk of injury. Contact surface of the obstacles must be cleaned with dry cloth if it is suspected to be wet. Check the serviceability of the obstacles to ensure they do not give way in between practice.

(e) **Qualified Supervision**. Obstacles should be attempted under the guidance of trained instructors to ensure proper technique and safety.

(f) **First Aid Preparedness**. Ensure that first aid facilities are readily available for any injuries that may occur.

17. **Benefits of Obstacle Course Training**.

(a) **Physical Fitness**. Improves overall strength, endurance, and cardiovascular health, making cadets physically robust.

(b) **Agility**. Sharpens the ability to move swiftly and fluidly, essential in navigating dynamic environments.

(c) **Mental Toughness**. Builds resilience and grit by pushing cadets to overcome difficult challenges.

(d) **Coordination and Balance**. Enhances coordination between the mind and body, resulting in better balance and precision in movement.

(e) **Risk-Taking Ability**. Encourages cadets to assess risks and face their fears, fostering confidence in decision-making.

(f) **Problem-Solving Skills**. Teaches cadets to quickly analyze situations and make effective decisions under pressure.

(g) **Team Spirit**. Strengthens collaboration and communication skills, fostering teamwork and unity among cadets.

(h) **Developing Physical and Mental Fitness**. Builds both physical and mental endurance, essential for overcoming obstacles and challenges.

(j) **Enhancing Leadership and Teamwork**. Instils leadership qualities and the importance of working as a cohesive team.

(k) **Building Confidence and Self-Esteem**. Boosts self-confidence by successfully overcoming physical and mental challenges.

(l) **Preparing for Real-Life Challenges**. Equips cadets with the skills to handle both physical and psychological challenges in life.

(m) **Cultivating Discipline and Resilience**. Promotes discipline and resilience, essential qualities for personal and professional growth.

(n) **Promoting Tactical Thinking**. Encourages strategic thinking and the ability to quickly devise solutions in complex situations.

18. **Demonstration**. Demonstrations will be conducted by trained cadets under the supervision of PI Staff. This live demonstration allows cadets to observe proper technique and safety practices before attempting the course themselves.

CONCLUSION

19. Obstacle course training is a cornerstone of NCC's comprehensive physical fitness regimen. By teaching cadets how to navigate physical challenges, the training develops not just physical fitness but also critical mental and emotional traits like patience, courage, and determination. Intensive obstacle training, particularly during camps like the Thal Sainik Camp, builds the foundation for a well-rounded and confident individual, capable of tackling any challenge with poise and skill. The obstacle course training is more than a physical exercise; it is a test of mental robustness, teamwork, and personal courage.

SUMMARY

- Obstacle training is a vital component of the NCC curriculum, designed to enhance cadets' physical fitness, mental resilience, and leadership skills. The training involves navigating a series of ten obstacles, such as the Straight Balance, High Wall, Double Ditch, and Zig-Zag Balance, each testing balance, agility, strength, and coordination.

- The primary objectives include building confidence, patience, and courage while improving agility, problem-solving abilities, and teamwork. Safety is a key focus, with cadets trained under supervision and proper techniques emphasized. The training fosters physical endurance, mental robustness, and a spirit of camaraderie.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. The Standard Obstacle Course that the NCC SW are required to cross consists of _____ obstacles?

(a) 6 (b) 8

(c) 10 (d) 12

Q2. Clear Jump has a bar that is placed _____above the ground?

(a) 1 ft (b) 2 ft

(c) 4 ft (d) 1.5 Ft

Q3. The Standard height of High Wall obstacles is ?

(a) 10ft (b) 8 ft

(c) 6 ft (d) 12 ft

Q4. Which of the following obstacle is optional for Girl cadet?

(a) Straight Balance (b) Clear Jump

(c) Zig Zag (d) High wall

Q5. What is the purpose of obstacle course training for NCC cadets?

(a) To enhance academic skills

(b) To develop musical talents

(c) To improve physical strength, confidence, courage, and willpower

(d) To practice marksmanship

Q6. How is the "Clear Jump" obstacle in the Standard Obstacle Course described?

(a) Jumping over a wooden slab with open arms

(b) Crossing a gate with parallel bars

(c) Running over a Zig-Zag structure with open hands

(d) Jumping over a straight bar without body contact

Q7. Which obstacle involves jumping over a bricked wall with plaster on both sides?

(a) Straight Balance (b) High Wall

(c) Double Ditch (d) Ramp

Q8. What safety measure is emphasized during the Obstacle Course training?

(a) Selecting cadets randomly

(b) Conducting training without any supervision

(c) Avoiding wet and slippery obstacles

(d) Ignoring the correct technique

Q9. What benefit does obstacle course training provide according to the lesson?

(a) Musical skills improvement (b) Flexibility and mental strength

(c) Enhancing academic performance (d) Learning marksmanship

Q10. In the "Left Hand Vault" obstacle, how does the cadet overcome it?

(a) Jumping over using the right hand (b) Jumping over using both hands

(c) Jumping over using the left hand (d) Crawling and climbing

Q11. Where does intensive training for NCC cadets take place, especially in Thal Sainik Camps?

(a) Obstacle Course centers (b) Regular classrooms

(c) Music training camps (d) PT dress training camps

Q12. According to the text, what qualities does obstacle course training aim to increase in NCC cadets?

(a) Laziness and indifference (b) Agility, courage, patience, and confidence

(c) Lack of teamwork (d) Fear and self-doubt

Q13. What is the primary focus during the conduct of Obstacle Course training, according to safety measures?

(a) Ignoring individual and team timings

(b) Emphasizing individual timings only

(c) Emphasizing team timings only

(d) Ensuring suitable and physically fit cadets are selected

Q14. How many obstacles does the Standard Obstacle Course consist of?

(a) Five (b) Ten

(c) Fifteen (d) Twenty

Q15. What is the structure of the "Clear Jump" obstacle?

(a) Zig-Zag like wooden bar (b) 6 feet high bricked wall

(c) 18 ft long straight bar (d) Wooden slab above ground level

Q16. How is the "High Wall" obstacle crossed?

(a) Running and jumping over it (b) Crawling under it

(c) Kicking the wall with both legs (d) Climbing over with hands and feet

Q17. Which safety measure is emphasized during the conduct of Obstacle Course training?

(a) Wet and slippery obstacles are encouraged

(b) Training without supervision

(c) Suitable and physically fit cadets only to be selected

(d) Obstacles done without correct techniques

Q18. What benefit does obstacle course training NOT provide?

(a) Mental strength (b) Physical fitness

(c) Artistic skills (d) Risk-taking ability

Q19. Where is intensive training given to NCC cadets, especially in Thal Sainik Camps?

(a) In regular classrooms (b) During outdoor sports events

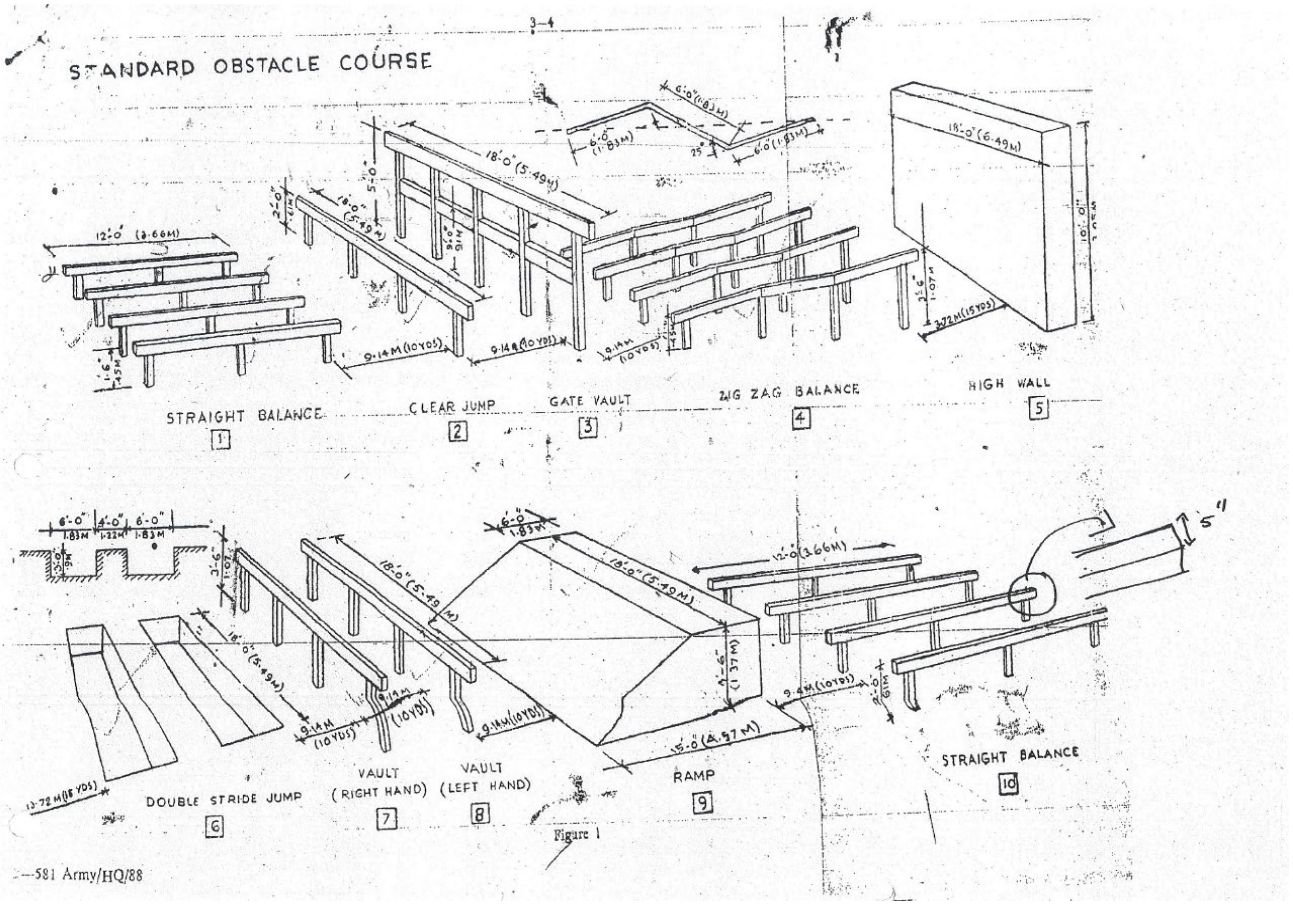
(c) In the obstacle course training area (d) In the camps, especially in Thal Sainik Camps

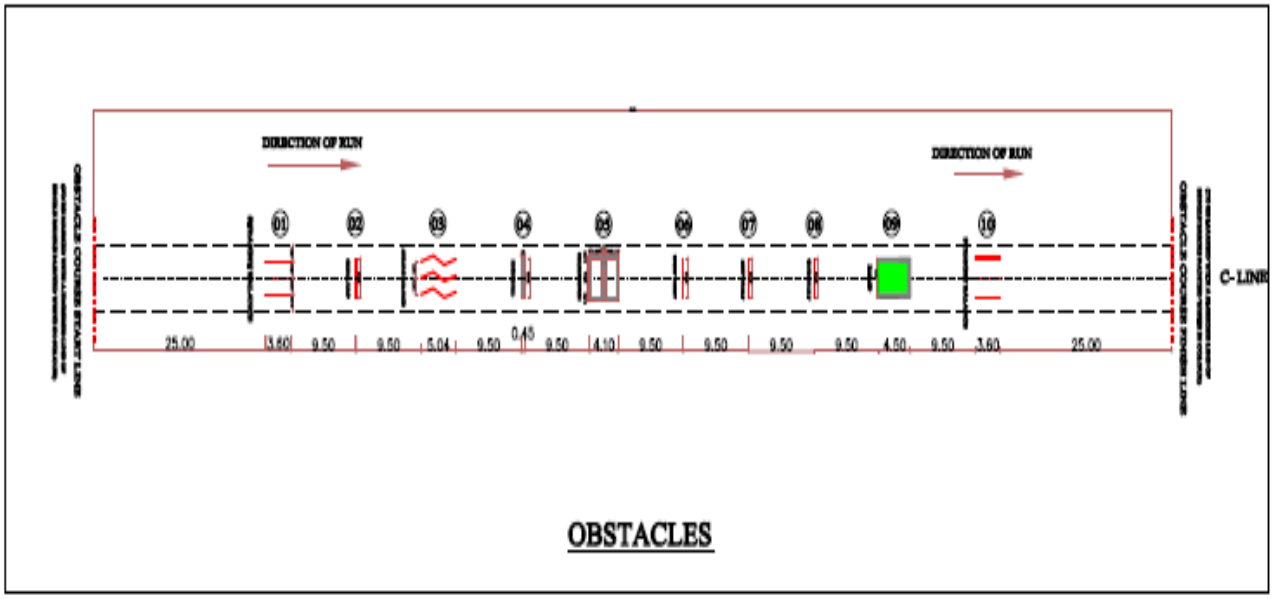
Q20. What is one of the benefits of completing the Obstacle Course training?

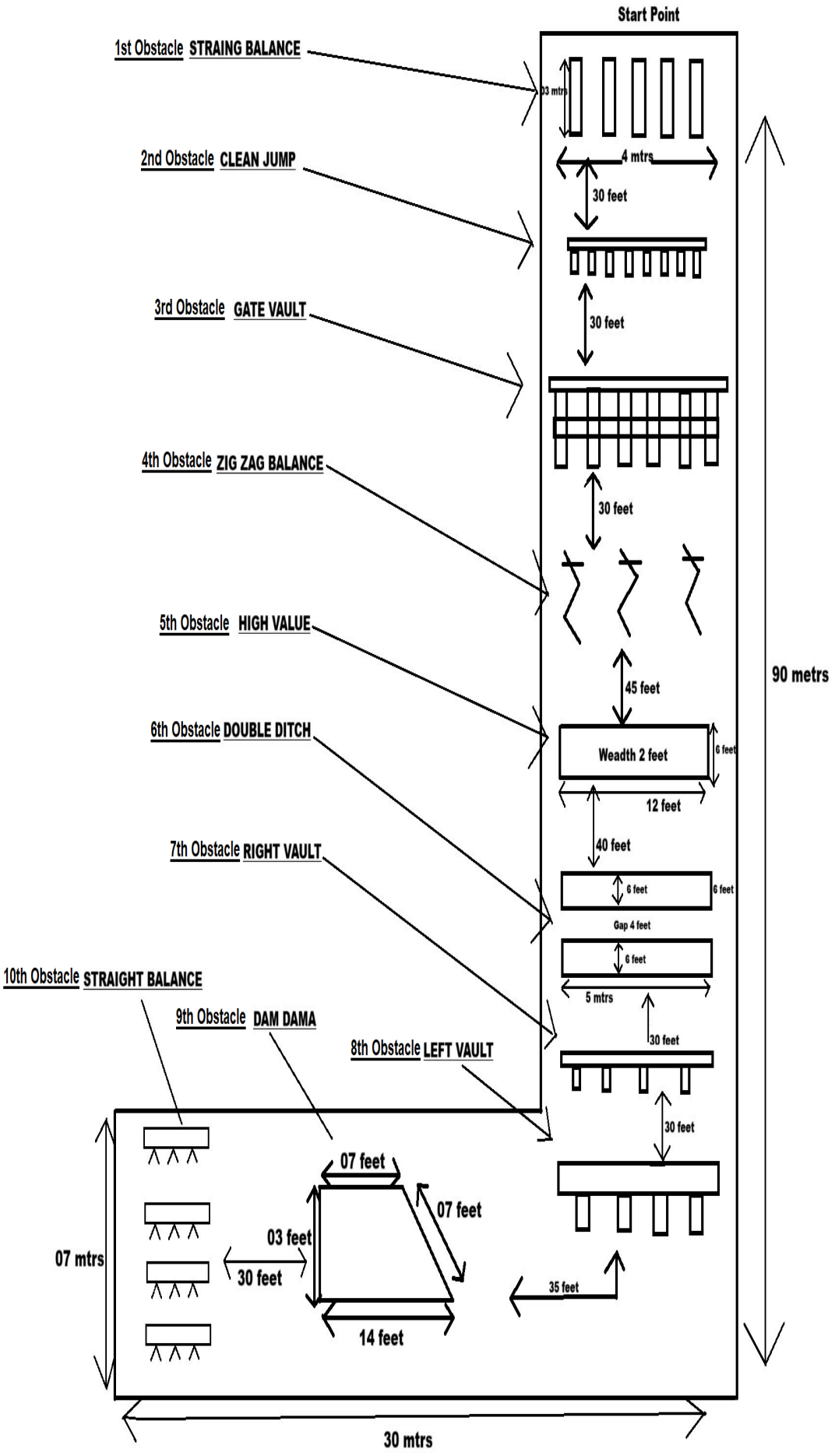
(a) Enhancing artistic skills (b) Improving risk-taking ability

(c) Developing culinary expertise (d) Mastering computer programming

SCHEMATIC LAYOUT OF STD OBST COURSE
(Basic and Battle Physical Training Pamphlet No 5 Obstacle Training - 1970)






Start Point
1st Obstacle STRAING BALANCE
03 mtrs
2nd Obstacle CLEAN JUMP
4 mtrs
30 feet
3rd Obstacle GATE VAULT
30 feet
4th Obstacle ZIG ZAG BALANCE
30 feet
5th Obstacle HIGH VALUE
45 feet
6th Obstacle DOUBLE DITCH
Weadth 2 feet
6 feet
12 feet
7th Obstacle RIGHT VAULT
40 feet
6 feet
6 feet
Gap 4 feet
6 feet
5 mtrs
30 feet
8th Obstacle LEFT VAULT
30 feet
30 feet
90 metrs
10th Obstacle STRAIGHT BALANCE
9th Obstacle DAM DAMA
07 feet
07 feet
03 feet
30 feet
07 mtrs
35 feet
14 feet
30 mtrs